Matemáticas B

**4º ESO**

SOLUCIONES DE LA FICHA · UNIDAD 3

# Ecuaciones e inecuaciones

FentMates

Intenta resolver cada ejercicio antes de mirar su solución. Puedes comprobar las ecuaciones y las inecuaciones con las herramientas interactivas de los apuntes de la unidad.

## 1 Ecuaciones de segundo grado

**1.** ★☆☆

**a)** $\boxed{x = \pm 4}$ **b)** $\boxed{x = 0,\ x = -4}$ **c)** $\boxed{x = \pm 5}$

**d)** No tiene solución. **e)** $\boxed{x = 0,\ x = \frac{3}{5}}$ **f)** $\boxed{x = \pm\frac{1}{2}}$

**2.** ★☆☆

**a)** $\boxed{x = 2,\ x = 3}$ **b)** $\boxed{x = 3,\ x = -5}$ **c)** $\boxed{x = 3,\ x = \frac{1}{2}}$

**d)** $\boxed{x = 3}$ (doble) **e)** No tiene solución ($\Delta < 0$). **f)** $\boxed{x = \frac{1}{2},\ x = -\frac{1}{3}}$

**3.** ★★☆

**a)** $x^2 + x - 6 = 0$: $\boxed{x = 2,\ x = -3}$ **b)** $x^2 - 5x - 14 = 0$: $\boxed{x = 7,\ x = -2}$

**c)** $x^2 + 5x - 6 = 0$: $\boxed{x = 1,\ x = -6}$ **d)** $2x^2 - 3x - 9 = 0$: $\boxed{x = 3,\ x = -\frac{3}{2}}$

**4.** ★★★

**a)** $x^2 + 9x - 22 = 0$: $\boxed{x = -11,\ x = 2}$

**b)** $x^2 - 4x + 3 = 0$: $\boxed{x = 1,\ x = 3}$

**c)** $x^2 - 5x + 4 = 0$: $\boxed{x = 1,\ x = 4}$

**d)** $x^2 + 3x - 10 = 0$: $\boxed{x = -5,\ x = 2}$

**e)** $x^2 - x - 6 = 0$: $\boxed{x = -2,\ x = 3}$

**f)** $x^2 - 2x + 3 = 0$: no tiene solución

**g)** $x^2 + 4x + 1 = 0$: $\boxed{x = -2 - \sqrt{3},\ x = -2 + \sqrt{3}}$

**h)** $x^2 - 2x - 6 = 0$: $\boxed{x = 1 - \sqrt{7},\ x = 1 + \sqrt{7}}$



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**5.** ★★☆

**a)** $\Delta = -56$: ninguna **b)** $\Delta = 0$: una (doble) **c)** $\Delta = 5$: dos

$\Delta = 36 - 4k = 0 \Rightarrow \boxed{k = 9}$.

**6.** ★★★ $(x-3)(x+5) = 0 \Rightarrow \boxed{x^2 + 2x - 15 = 0}$ (o cualquier múltiplo).

## 2 Ecuaciones bicuadradas

**7.** ★☆☆

**a)** $t = 9, t = 1$: $\boxed{x = \pm 3, x = \pm 1}$ **b)** $t = 9, t = -4$: $\boxed{x = \pm 3}$

**c)** $t = -1, t = -2$: no tiene solución. **d)** $t = 4, t = \frac{1}{4}$: $\boxed{x = \pm 2, x = \pm \frac{1}{2}}$

**8.** ★★☆

**a)** $x^2(x^2 - 8) = 0$: $\boxed{x = 0, x = \pm 2\sqrt{2}}$

**b)** $x^4 + x^2 - 12 = 0$: $\boxed{x = \pm\sqrt{3}}$

**c)** $\boxed{x = \pm 2}$

**9.** ★★★

**a)** $x^4 - 3x^2 - 4 = 0$: $\boxed{x = -2, x = 2}$

**b)** $x^4 - 5x^2 + 4 = 0$: $\boxed{x = -2, x = -1, x = 1, x = 2}$

**c)** $x^4 - x^2 - 12 = 0$: $\boxed{x = -2, x = 2}$

**d)** $x^4 - 10x^2 + 16 = 0$: $\boxed{x = -2\sqrt{2}, x = -\sqrt{2}, x = \sqrt{2}, x = 2\sqrt{2}}$

**e)** $2x^4 - 9x^2 + 9 = 0$: $\boxed{x = -\sqrt{3}, x = -\frac{\sqrt{6}}{2}, x = \frac{\sqrt{6}}{2}, x = \sqrt{3}}$

**f)** $4x^2 - 9 = 0$: $\boxed{x = -\frac{3}{2}, x = \frac{3}{2}}$

## 3 Ecuaciones de grado mayor que 2

**10.** ★★☆

**a)** $\boxed{x = 1, x = 2, x = 3}$ **b)** $\boxed{x = 1, x = -1, x = -2}$

**c)** $(x-1)^2(x+2) = 0$: $\boxed{x = 1, x = -2}$ **d)** $\boxed{x = 2, x = -\frac{1}{2}, x = -2}$

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**11.** ★★☆

**a)** $(x-1)(x+1)(x+2)(x-3)=0$: $\boxed{x=\pm 1,\ x=-2,\ x=3}$

**b)** $x(x+2)(x-2)=0$: $\boxed{x=0,\ x=\pm 2}$

**c)** $x(x+1)^2(x-1)=0$: $\boxed{x=0,\ x=-1,\ x=1}$

## 4 Ecuaciones factorizadas

**12.** ★☆☆

**a)** $\boxed{x=4,\ x=-1}$

**b)** $\boxed{x=0,\ x=-\frac{3}{2},\ x=5}$

**c)** $\boxed{x=1,\ x=-7}$

**d)** $\boxed{x=-2,\ x=\frac{1}{4}}$

**13.** ★★☆

**a)** $\boxed{x=\pm 4,\ x=-3}$

**b)** $\boxed{x=1,\ x=2}$ ($x^2+5=0$ no da ninguna)

**c)** $\boxed{x=0,\ x=4,\ x=-2}$

**d)** No es «$=0$»: $x^2-x-6=0$, $\boxed{x=3,\ x=-2}$

## 5 Ecuaciones con fracciones algebraicas

**14.** ★☆☆

**a)** $\boxed{x=1}$

**b)** $\boxed{x=3}$

**c)** $x^2-2x-1=0$: $\boxed{x=1\pm\sqrt{2}}$

**d)** $x^2-5x+4=0$: $\boxed{x=1,\ x=4}$

**15.** ★★☆

**a)** $x^2-3x-10=0\Rightarrow x=5,\ x=-2$; $x=-2$ anula un denominador: $\boxed{x=5}$

**b)** $x+5=6\Rightarrow x=1$, que anula $x-1$: $\boxed{\text{no tiene solución}}$

**c)** $2x^2-3x-2=0$: $\boxed{x=2,\ x=-\frac{1}{2}}$

**d)** $x^2+4=8\Rightarrow x=\pm 2$, y las dos anulan un denominador: $\boxed{\text{no tiene solución}}$

**e)** $x^2-2x-1=0$: $\boxed{x=1\pm\sqrt{2}}$

**f)** $2(x+1)-3=x$: $\boxed{x=1}$

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**16.** ★★★ m.c.m. $= 6x(x+1)$: $6(x+1)^2 - 6x^2 = 5x(x+1) \Rightarrow 5x^2 - 7x - 6 = 0 \Rightarrow \boxed{x = 2,\ x = -\frac{3}{5}}$ (las dos valen).

## 6 Ecuaciones con radicales

**17.** ★☆☆

**a)** $\boxed{x = 4}$ **b)** $\boxed{x = 13}$ **c)** $\boxed{x = \pm 4}$ **d)** No tiene: una raíz no es negativa.

**18.** ★★☆

**a)** $x^2 - x - 2 = 0 \Rightarrow x = 2,\ x = -1$; $x = -1$ no vale ($\sqrt{1} \neq -1$): $\boxed{x = 2}$

**b)** $x^2 - 15x + 50 = 0 \Rightarrow x = 5,\ x = 10$; $x = 10$ no vale ($10 + 3 \neq 7$): $\boxed{x = 5}$

**c)** $x^2 - 3x - 4 = 0 \Rightarrow x = 4,\ x = -1$; $x = -1$ no vale ($1 + 1 \neq 0$): $\boxed{x = 4}$

**d)** $x^2 - 10x + 9 = 0 \Rightarrow x = 1,\ x = 9$; $x = 1$ no vale ($2 \neq -2$): $\boxed{x = 9}$

**e)** $x^2 - 2x - 3 = 0 \Rightarrow x = 3,\ x = -1$; $x = -1$ no vale: $\boxed{x = 3}$

**f)** $x^2 - 5x - 6 = 0 \Rightarrow x = 6,\ x = -1$; $x = -1$ no vale ($-1 - 3 \neq 2$): $\boxed{x = 6}$

**g)** $x^2 - x = 0 \Rightarrow x = 0,\ x = 1$; las dos valen: $\boxed{x = 0,\ x = 1}$

**h)** $x^2 - 7x + 10 = 0 \Rightarrow x = 5,\ x = 2$; $x = 2$ no vale ($3 + 1 \neq 2$): $\boxed{x = 5}$

**19.** ★★★

**a)** $x + 7 = 1 + 2\sqrt{x} + x \Rightarrow \sqrt{x} = 3$: $\boxed{x = 9}$

**b)** $x - 12 = -8\sqrt{x-3} \Rightarrow x^2 - 88x + 336 = 0 \Rightarrow x = 4,\ x = 84$; $x = 84$ no vale ($13 + 9 \neq 4$): $\boxed{x = 4}$

## 7 Ecuaciones exponenciales

**20.** ★☆☆

**a)** $\boxed{x = 5}$ **b)** $\boxed{x = 5}$ **c)** $\boxed{x = -1}$

**d)** $\boxed{x = \frac{3}{2}}$ **e)** $\boxed{x = \pm 2}$ **f)** $\boxed{x = -4}$

**21.** ★☆☆

**a)** $\boxed{x = \frac{\log 10}{\log 2} \approx 3{,}322}$ **b)** $\boxed{x = \frac{\log 50}{\log 3} \approx 3{,}561}$

**c)** $\boxed{x = 1 + \frac{\log 30}{\log 7} \approx 2{,}748}$ **d)** $\boxed{x = \frac{\log 2}{\log 1{,}05} \approx 14{,}207}$

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**22.** ★★☆

**a)** $3 \cdot 2^x = 24$: $\boxed{x = 3}$ **b)** $8 \cdot 3^x = 72$: $\boxed{x = 2}$ **c)** $\frac{6}{5} \cdot 5^x = 30$: $\boxed{x = 2}$

**23.** ★★★

**a)** $t = 2^x$: $t = 1, t = 4$; $\boxed{x = 0, x = 2}$

**b)** $t = 3^x$: $t = 1, t = 9$; $\boxed{x = 0, x = 2}$

**c)** $t = 4$ ($t = -3$ no vale): $\boxed{x = 2}$

## 8 Ecuaciones logarítmicas

**24.** ★☆☆

**a)** $\boxed{x = 1000}$ **b)** $\boxed{x = 15}$ **c)** $\boxed{x = 1}$ **d)** $\boxed{x = \frac{26}{3}}$

**25.** ★★☆

**a)** $\boxed{x = 5}$

**b)** $\boxed{x = 3}$

**c)** $x^2 - 5x + 6 = 0$: $\boxed{x = 2, x = 3}$

**d)** $x^2 + 9x - 10 = 0$; $x = -10$ no vale: $\boxed{x = 1}$

**e)** $x^2 - x - 2 = 10 \Rightarrow x = 4, x = -3$; $x = -3$ no vale: $\boxed{x = 4}$

**f)** $x^2 = 9(x-2) \Rightarrow x^2 - 9x + 18 = 0$; las dos valen: $\boxed{x = 3, x = 6}$

**g)** $2x + 4 = 3x$: $\boxed{x = 4}$

**h)** $(x+2)(x-4) = 27 \Rightarrow x^2 - 2x - 35 = 0$; $x = -5$ no vale: $\boxed{x = 7}$

**26.** ★★★

**a)** $x + 6 = 10(x - 3)$: $\boxed{x = 4}$

**b)** $x^2 - 2x - 8 = 0$; $x = -2$ no vale: $\boxed{x = 4}$

## 9 Inecuaciones de primer grado

**27.** ★☆☆

**a)** $\boxed{(4, +\infty)}$




**b)** $\boxed{[-2, +\infty)}$




**c)** $\boxed{(-\infty, -4)}$




**d)** $\boxed{(-\infty, -4]}$




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**28.** ★★☆

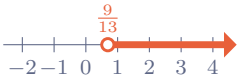
**a)** $0 \geq 0$, siempre: $\boxed{\mathbb{R}}$




**b)** $\boxed{(-\infty, 3)}$




**c)** $\boxed{(\frac{9}{13}, +\infty)}$




**d)** $-1 > 5$, nunca: $\boxed{\varnothing}$

**29.** ★★☆ $15 + 0{,}5x > 40 \Rightarrow x > 50$: con $\boxed{\text{más de 50 GB}}$ es más cara la B.

## 10 Inecuaciones de segundo grado o superior

**30.** ★★☆

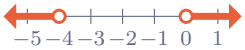
**a)** $\boxed{(-3, 3)}$




**b)** $\boxed{(-\infty, -2] \cup [5, +\infty)}$




**c)** $\boxed{(-\infty, -4) \cup (0, +\infty)}$




**d)** $\boxed{(-1, 3)}$




**e)** Solo $x = 2$: $\boxed{\{2\}}$




**f)** $\Delta < 0$ y $a > 0$, siempre positivo: $\boxed{\mathbb{R}}$




**31.** ★★★

**a)** $\boxed{(-1, 0) \cup (1, +\infty)}$




**b)** $\boxed{(-\infty, -2] \cup [1, 4]}$




**c)** $x(x+1)(x-3) \geq 0$: $\boxed{[-1, 0] \cup [3, +\infty)}$




**32.** ★★☆ $x(x+3) < 40 \Rightarrow x^2 + 3x - 40 < 0 \Rightarrow x \in (-8, 5)$. Como $x > 0$: $\boxed{x \in (0, 5)}$.

## 11 Problemas

**33.** ★★☆ $x(x+1) = 156 \Rightarrow x^2 + x - 156 = 0 \Rightarrow x = 12$ ($x = -13$ no es natural). Son $\boxed{12 \text{ y } 13}$.

**34.** ★★☆ $x + y = 17$ y $x^2 + y^2 = 169$: $x^2 + (17 - x)^2 = 169 \Rightarrow x^2 - 17x + 60 = 0 \Rightarrow$ $\boxed{12 \text{ cm y } 5 \text{ cm}}$.

**35.** ★★☆ $5000 \cdot 1{,}03^t = 6000 \Rightarrow t = \dfrac{\log 1{,}2}{\log 1{,}03} \approx 6{,}17$: hacen falta $\boxed{7 \text{ años}}$.

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**36.** ★★☆ $\frac{600}{x} - \frac{600}{x+2} = 25 \Rightarrow x^2 + 2x - 48 = 0 \Rightarrow x = 6$ ($x = -8$ no tiene sentido): son $\boxed{6 \text{ amigos}}$.

**37.** ★★★ $\frac{1}{x} + \frac{1}{x+5} = \frac{1}{6} \Rightarrow x^2 - 7x - 30 = 0 \Rightarrow x = 10$ ($x = -3$ no tiene sentido): $\boxed{10 \text{ h y } 15 \text{ h}}$.

**38.** ★★☆ $x + \sqrt{x} = 42 \Rightarrow \sqrt{x} = 42 - x \Rightarrow x^2 - 85x + 1764 = 0 \Rightarrow x = 36, x = 49$; $x = 49$ no vale ($49 + 7 \neq 42$): $\boxed{36}$.

**39.** ★★☆ $x + \frac{1}{x} = \frac{26}{5} \Rightarrow 5x^2 - 26x + 5 = 0$: $\boxed{x = 5 \text{ o } x = \frac{1}{5}}$.

**40.** ★★★ $x^2 < 3x \Rightarrow x(x-3) < 0$: $\boxed{(0, 3)}$




$x^3 > x \Rightarrow x(x-1)(x+1) > 0$: $\boxed{(-1, 0) \cup (1, +\infty)}$




**41.** ★★★ $2^{t/3} = 32 = 2^5 \Rightarrow \frac{t}{3} = 5 \Rightarrow \boxed{t = 15 \text{ horas}}$.

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